

SIH26036 - Development of an Online Verification System for Weighing and Measuring Instruments

Overview: Digitizes the Legal Metrology verification workflow for weighing and measuring instruments.

Problem: Manual applications, paper records, delayed inspections, fake certificates, and lack of tracking.

Solution: Online portal for registration, verification requests, inspector assignment, inspection reports, QR-code based digital certificates, renewal reminders, and dashboards.

Modules: Registration, Instrument Management, Verification Requests, Inspector Dashboard, Certificate Generation, QR Verification, Notifications, Reports.

Users: Business Owner, Inspector, Administrator, Public.

Tech Stack: React/Flutter, FastAPI or Django (Python), PostgreSQL/MySQL, JWT, QR Code library, ReportLab.

Physical Components: Laptop, Android phone, Internet, Smartphone camera, QR labels. Demo weighing scale optional.

Ratings:

Difficulty: ★★███

Impact: ★★★★★

Python Friendly: ★★★★★

Beginner Friendly: ★★★★★

Innovation: AI scheduling, fraud detection, predictive renewals, analytics dashboards.

Outcome: Faster verification, transparent certification, reduced paperwork, improved governance.